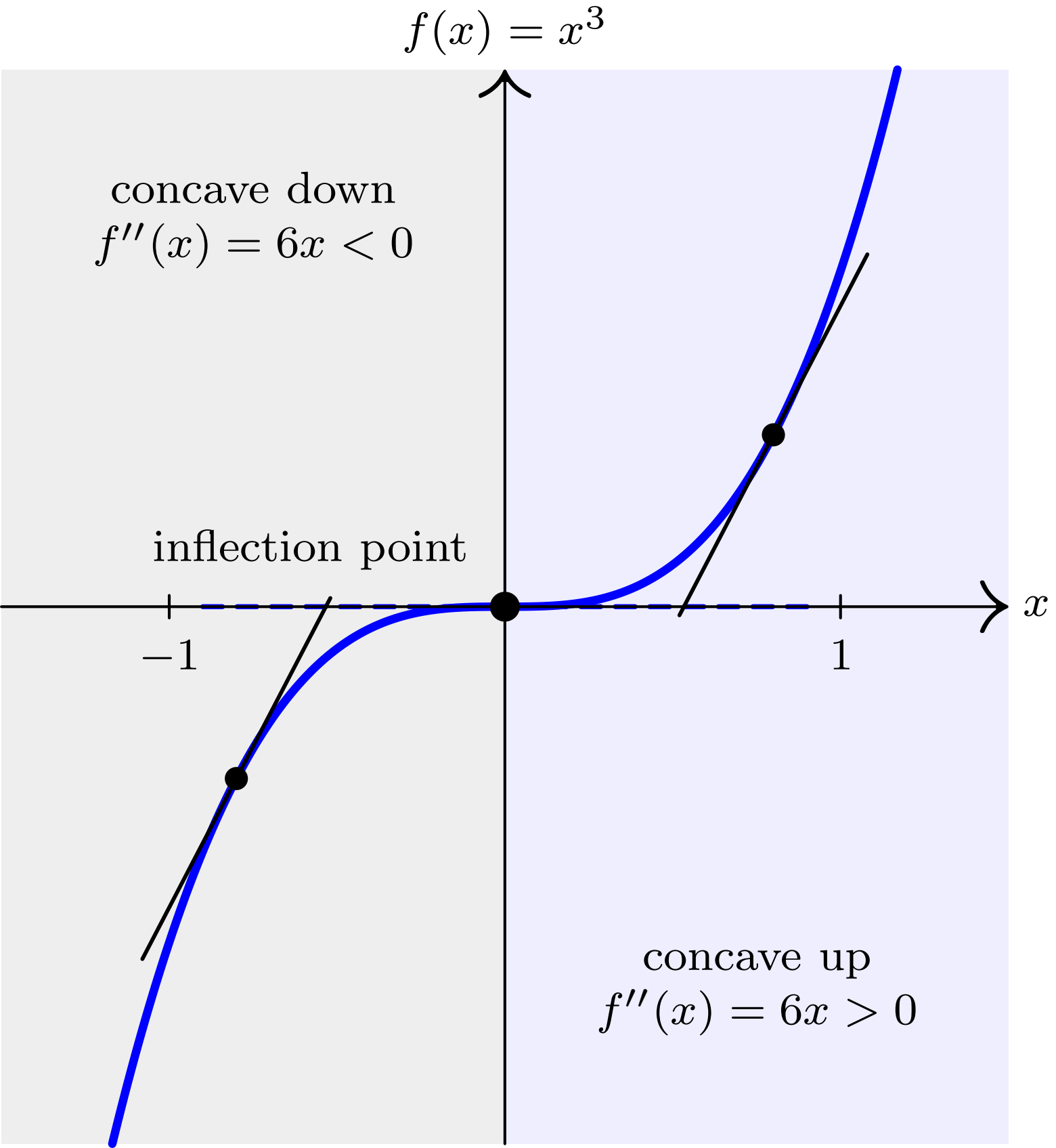
$f(x) = x^3$
concave down
$f''(x) = 6x < 0$
inflection point
-1
1
x
concave up
$f''(x) = 6x > 0$